

PIXEL ART IS A PALETTE, NOT A DOWNSCALE: REBUILDING TARGETS AND DECODING FOR 12–24 PX SPRITE GENERATION

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ABSTRACT

Text-to-image models fail at the smallest images people draw on purpose: 12–24 px RGBA game sprites, where a handful of colours and hard alpha edges *are* the medium. We show that the usual failure is not a modelling failure but a target failure. In the standard sprite corpus only 7% of the images are natively 16 px or smaller; the rest is 25–64 px artwork box-downscaled into the low-resolution buckets, so a model trained on it learns soft, fifty-colour blobs—and the evaluation reference inherits the same bias, rewarding exactly that. We rebuild the targets (drop 4,888 pixel-identical duplicates that leak 15.5% of the held-out set into training, forbid downscaling by more than $1.5\times$, add 14k natively small sprites from three permissively licensed corpora) and we evaluate against a reference drawn only from real low-resolution pixel art. Under this protocol the data fix alone takes FD-DINOv2 from 167.2 to 117.9 at 16 px. We then close the remaining gap where it actually is: real sprites use a median of six opaque colours while every sampling configuration we tested emits more than fifty. Projecting the predicted x_0 onto a per-sprite k -colour palette during the late denoising steps—so that the remaining steps repair the quantisation seams—reaches 31.8 ± 0.5 , better than quantising the finished samples (41.6), and $2.6\times$ better than the strongest external system under the same palette post-process (50.8 versus 134.4 at $n=200$); it also beats the mixed-provenance “real” sprites that the downscale-biased protocol treats as ground truth (77.0). We report an equally instructive negative result: cross-resolution self-guidance and its internalised one-pass variant, which win by 44% under the downscale-biased reference (6.24 against 11.10), give no gain once the reference is real pixel art, while independent-condition guidance transfers and composes with palette projection.

1 INTRODUCTION

Game sprites are among the smallest images anyone draws on purpose: an item or a character on a 12–24 px RGBA canvas, built from hard edges, a handful of flat colour regions and a transparent background. At these sizes there is no texture in which to hide a mistake. One wrong pixel changes the silhouette, and one blended colour reads as blur. The obvious recipe—prompt a large text-to-image model and shrink its output, or run a pixelization method on it—produces something recognisable but not a sprite: the result keeps the soft shading and the many colours of an illustration (Figure 1).

This paper is about a failure that sits one level below the model. We train a dedicated generator for this medium—a 72.5M-parameter pixel-space UNet that denoises RGBA sprites directly, conditioned on a frozen CLIP text encoder and on a *resolution bucket* label—and we find that its weaknesses are inherited from what it is asked to imitate. In the sprite corpus that work in this area uses, only 7.1% of the images are natively 16 px or smaller; two thirds are 25–64 px artwork that the pipeline box-downsamples into the low-resolution buckets. A model fitted to those targets learns exactly what they are: soft, fifty-colour blobs. Worse, the standard evaluation reference is drawn from the same pool, so it rewards the blur it caused: a sampler tuned on it prefers heavy guidance, and the ranking of methods under that reference does not survive contact with real pixel art.



Figure 1: Eight held-out captions at 16 px. Top: our model with ICG and palette projection ($k=4$); second row: the same model without the projection; then gpt-image-2 and FLUX.2-klein + pixel-art LoRA, both generated large and reduced by the same code, with the same four-colour post-process applied to gpt-image-2; bottom: the real held-out sprite. The external pipelines often produce the more attractive picture while being $2.6\times$ further from native pixel art (Table 5); our samples share the grammar — and the crudeness — of the data.

We therefore change both the targets and the yardstick. On the data side we (i) remove 4,888 pixel-identical duplicates, which had left a pixel-identical twin in training for 15.5% of the held-out evaluation sprites, (ii) forbid feeding a sprite to a bucket more than $1.5\times$ below its native size, and (iii) add 14,999 natively small sprites cut from three permissively licensed OpenGameArt collections and captioned with a vision–language model. On the evaluation side we define the *native protocol*: FD-DINOv2 against real sprites that are natively at most R px, never downscaled. At 16 px the data work alone moves the model from 167.2 to 117.9 under this protocol.

What remains is a property of the medium that continuous diffusion has no way to express. Real 16 px sprites use a median of six opaque colours; every sampling configuration we tested—classifier-free guidance at any weight, cross-resolution self-guidance, independent-condition guidance—emits more than fifty, spending its extra colours on $\pm$ few-level dither inside regions that should be flat. We close that gap inside the sampler: during the late denoising steps we project the predicted x_0 onto a per-sprite k -colour palette (batched k -means over the opaque pixels, alpha untouched) and continue denoising, so the remaining steps repair the seams that hard quantisation leaves. At 16 px this takes FD from 87.2 to 31.8 ± 0.5 ; quantising the finished samples instead only reaches 41.6.

Finally, we report a negative result that we consider as informative as the positive one. Cross-resolution self-guidance—querying the same network under a lower bucket label and extrapolating away from that prediction—and its internalised one-pass variant were, on the downscale-biased reference, our strongest method (a 44% improvement over the best-tuned CFG). Under the native protocol they give nothing (114.9 and 117.0 versus 113.2 for unguided sampling), and the internalised variant is actively harmful (170.3). Independent-condition guidance, an external method, transfers



Figure 2: All four rows are 16 px RGBA sprites on the same checkerboard, nearest-neighbour magnified. **Row 1:** real sprites that were drawn at this size — a median of six opaque colours and hard alpha edges. **Row 2:** what the corpus feeds into its 16 px bucket, which is 25–64 px artwork box-downscaled, at 49 colours; 92.9% of the conventional evaluation reference looks like this. **Rows 3–4** share the caption of their column: gpt-image-2 rendered at 1024 px and reduced (111 colours), and our model with palette-projected sampling (4 colours). Row 3 is the easiest on the eye and the furthest from the medium; the paper is about that gap and about the reference set that hides it.

and composes with palette projection. Which reference set one scores against decides which of these conclusions one publishes.

Contributions.

- A diagnosis of the targets used for tiny-sprite generation: native-resolution statistics, duplicate leakage, and the resulting bias of the standard evaluation reference; and a rebuilt corpus plus the *native protocol* that removes both (Section 3).
- **Palette-projected sampling:** a projection of the predicted x_0 onto a per-sprite k -colour palette in the late denoising steps, which is where the remaining distance to real pixel art actually lies. It lowers FD-DINOv2 at 16 px from 87.2 to 31.8 ± 0.5 (3 seeds), beats post-hoc quantisation (41.6), and costs 1.3 points of CLIP retrieval at $k=6$ (Section 4).
- An external comparison in which every system receives the same palette post-process: at 16/20/24 px we reach 50.8/152.5/192.2 against 134.4/448.6/478.0 for gpt-image-2 with downscaling, 150.1/473.9/472.4 for FLUX.2-klein with a pixel-art LoRA, and 244.2/526.3/527.2 for SDXL with Pixel Art XL (Section 5.5).
- A negative result with its mechanism: cross-resolution guidance and its internalisation win by 40% under a downscale-biased reference and vanish under a native one; we give the per-image statistics (palette size, flat-region fraction, contrast) that explain both outcomes (Section 6).

2 RELATED WORK

Guidance references. CFG (Ho & Salimans, 2022) extrapolates the conditional prediction away from an unconditional one. Many variants keep that reference and reshape the update: guidance intervals (Kynkäänniemi et al., 2024), APG (Sadat et al., 2025a), frequency decoupling (Sadat et al., 2025c), CADs (Sadat et al., 2024), CFG++ (Chung et al., 2025) and feedback or stage-wise schedules (Koulischer et al., 2025; Jin et al., 2026). A second line replaces the reference. Autoguidance (Karras et al., 2024) uses a smaller or less-trained version of the same model. Training-free weak predictions come from perturbed self-attention (PAG, SAG, SEG; Ahn et al., 2024; Hong et al., 2023; Hong, 2024), sliding windows (Adaloglou et al., 2025), stochastic sub-networks (Chen et al., 2026), a random condition or a perturbed time embedding (ICG, TSG; Sadat et al., 2025b), or a semantically degraded text condition (CDG; Han et al., 2026). Other weak predictions come from

auxiliary or internal heads (Zhou et al., 2026; Yuan et al., 2026; Fu et al., 2026). CDG is closest in spirit: it too degrades one condition of the same network. It degrades the *text*, whereas we degrade the *resolution label* and keep the caption fixed. We run CDG, ICG, TSG, SEG and PAG under our protocol (Table E).

Guidance without a second pass. Guidance distillation (Meng et al., 2023) regresses a student onto a guided teacher. Model guidance (Tang et al., 2025) and guidance-free training (Chen et al., 2025) instead change the training objective so that its fixed point is the guided prediction. GFT does this with a stop-gradient unconditional reference, a scalar β input, and the true noise as the target, and reports parity with CFG. We use GFT’s objective unchanged and swap its reference. Our β input reuses the guidance-scale embedding of latent consistency models (Luo et al., 2023).

Resolution conditioning. Multi-resolution training with a size or bucket label is standard: aspect-ratio bucketing (Anlatan (NovelAI), 2022), SDXL’s micro-conditioning (Podell et al., 2024), Matryoshka diffusion (Gu et al., 2024), FiT (Lu et al., 2024) and simple diffusion (Hoogetboom et al., 2023). SDXL documents that a small `original_size` reproduces the blur of low-resolution training images, and common practice passes it to the negative branch of CFG (von Platen et al., 2022). We test this as a reference in Appendix I and find no benefit. To our knowledge, no prior work uses a model’s own lower-resolution label as a guidance reference or internalizes one.

Palettes and quantisation. Reducing an image to a small palette is classical: median cut (Heckbert, 1982), error diffusion (Floyd & Steinberg, 1976) and, for pixel art specifically, joint palette-and-pixel optimisation (Gerstner et al., 2012) and palette-conditioned generation (Li, 2025). In diffusion, hard constraints are usually enforced by projecting the predicted x_0 at each step, as in plug-and-play priors and inverse-problem solvers (Venkatakrishnan et al., 2013; Chung et al., 2023), and by quantisation-aware or discrete-state formulations (Austin et al., 2021; Gu et al., 2022). Our projection is the same device applied to the property that defines this medium, with the palette estimated per sprite from the model’s own current belief rather than supplied as a condition; unlike a discrete-state model it needs no change of training and keeps the continuous denoiser.

Data and protocol audits. Reports that a benchmark’s reference set decides the ranking are common in generative evaluation: FID’s sensitivity to the feature extractor and to preprocessing (Parmar et al., 2022; Stein et al., 2023), duplicate and near-duplicate contamination in image corpora (Barz & Denzler, 2020; Webster et al., 2023), and dataset-bias analyses more generally (Torralba & Efros, 2011). Our contribution here is narrow and concrete: for tiny sprites, the conventional reference is dominated by downsampled artwork, and that single fact reverses the ranking of the guidance methods we and others would otherwise report.

Pixel art. Most pixel-art work converts an existing image, by optimisation (Gerstner et al., 2012), contrast-aware downscaling (PixelOE; Yeh, 2024) or learned pixelization (Han et al., 2018; Wu et al., 2022). Generative approaches include GANs for character sprites (Coutinho & Chaimowicz, 2022), pose-conditioned sprite sheets (Hsieh et al., 2024), a palette-conditioned small DDPM (Li, 2025), SD- π XL, which optimises one image by score distillation from SDXL (Binnering & Sorkine-Hornung, 2024), and community LoRAs such as Pixel Art XL (nerijs, 2023). We compare against conversion pipelines applied to strong generators; SD- π XL costs about 8 GPU-hours per sprite (Section 5.5). For evaluation we follow Stein et al. (2023) and use DINOv2 features (Oquab et al., 2024) rather than Inception (Heusel et al., 2017; Parmar et al., 2022).

3 THE TARGETS AND THE PROTOCOL

What the corpus actually contains. Work on tiny sprites uses OpenGameArt (OpenGameArt.org, 2025) together with the Kenney and Liberated Pixel Cup packs. After cutting sheets into connected components at their native scale and filtering out glyphs and UI, our copy holds 37,489 unique sprites. Their native sizes, however, are not tiny: only 710 are at most 12 px, 2,675 at most 16 px (7.1%) and 12,516 at most 24 px, while two thirds are 25–64 px artwork. A resolution-bucketed model is usually trained by box-downsampling every sprite into each lower bucket, so 92.9% of what the 16 px bucket sees is squashed artwork rather than pixel art drawn at 16 px. Only 4.6% of

the corpus is integer-upscaled pixel art that could be recovered by dividing by its block size, so the softness cannot be undone; it has to be avoided.

Duplicate leakage. Hashing every sprite by its exact pixel content reveals 4,888 duplicate copies (9.3% of the corpus). Because held-out sets are defined by path, a pixel-identical twin of a held-out sprite stays in training: 15.5% of the 3,000 held-out evaluation sprites, and 17.0% of the 200 used for the external comparison, are memorisable in this way.

Rebuilt targets. We (i) drop every duplicate copy and every copy of a held-out sprite, (ii) allow a sprite to feed a bucket only if it is at most $1.5\times$ larger than that bucket, and (iii) add natively small sprites: three further OpenGameArt licence collections (CC-BY-4.0, CC-BY-SA-3.0, OGA-BY-3.0) yield 37,226 cut sprites, of which 14,999 survive the same filters, near-duplicate collapse of animation frames and a cap of 60 sprites per source sheet. We caption them with gemini-3.8-flash on numbered image grids, keeping 13,987 captions (99.6% distinct, median 13 words); the provenance and licence of every sprite is recorded. The rebuilt corpus has 86,290 training rows and twice as many natively ≤ 16 px sprites as before. Appendix A gives the per-step counts and the verification sheets.

The native protocol. FD-DINOv2 (Stein et al., 2023) is computed against a reference set of real sprites pushed through the model’s own preprocessing. The usual choice—a random sample of the corpus—inherits the bias above. We therefore report the *native protocol*: the reference at resolution R is drawn only from sprites whose native size is at most R , so no downscaling ever touches it (1,337 reference and 1,338 held-out images at 16 px, real-vs-real floor 4.47). We report the biased protocol as well, because the two disagree systematically (Section 6), and we mark which is which in every table.

4 METHOD

4.1 A RESOLUTION-BUCKETED PIXEL-SPACE SPRITE MODEL

We generate RGBA sprites on an $R \times R$ canvas, $R \in \{12, 16, 20, 24\}$, from a caption c . The denoiser $\epsilon_\theta(x_t, t, c, b)$ is a UNet (diffusers UNet2DConditionModel; 72.5M parameters; channel widths 128/256/512; two cross-attention stages at R and $R/2$; no attention at $R/4$ except the mid block). It operates directly on the 4-channel image. Text enters through cross-attention to the 77 token states of a frozen CLIP ViT-B/32 text encoder (Radford et al., 2021). The resolution bucket $b \in \mathcal{B} = \{12, 16, 20, 24, 32, 48, 64\}$ enters as a class embedding added to the timestep embedding, and it is never dropped. We train with ϵ -prediction (Ho et al., 2020), a cosine schedule with $T = 1000$, AdamW at a constant learning rate of $5 \cdot 10^{-5}$, EMA 0.999 and 10% caption dropout. We sample with 100 ancestral DDPM steps from the EMA weights. The main model, **v8n**, is trained from scratch for 80k steps on the rebuilt corpus of Section 3; **v8f2** is the same run without the new sprites and **v7r** is the same architecture on the original targets, used as controls.

4.2 PALETTE-PROJECTED SAMPLING

A 16 px sprite is defined as much by how few colours it uses as by its shape: in the native reference set the median sprite has six opaque colours (12 px: 5; 20 and 24 px: 7) and 22.6% use at most four. A continuous diffusion model in RGB has no mechanism that snaps a region to a single value, and none of our samplers does: measured on 600 samples, plain CFG emits 56.2 colours and the best guidance rule 50.9, with the fraction of adjacent opaque pixel pairs that are exactly equal at 0.36–0.38 against 0.46 for real sprites. The extra colours are dither of a few levels inside regions that the model has, in effect, already decided are flat.

We remove that dither inside the sampler. At a step with $t/T \leq f$ we form the current estimate $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_t} \tilde{e}) / \sqrt{\bar{\alpha}_t}$, cluster its *opaque* RGB pixels into k groups with a batched k -means (8 iterations, deterministic initialisation at evenly spaced luminance quantiles), replace every pixel by its centroid, leave transparent pixels and the alpha channel untouched, and map the result back to a noise prediction,

$$\hat{x}_0^{\text{pal}} = \Pi_k(\hat{x}_0), \quad \tilde{e} \leftarrow (x_t - \sqrt{\bar{\alpha}_t} \hat{x}_0^{\text{pal}}) / \sqrt{1 - \bar{\alpha}_t}, \quad (1)$$



Figure 3: Palette-projected sampling at 16 px, same prompts and seed. Without projection (top) flat regions carry a few levels of dither; projecting the predicted x_0 onto four or six colours during the late steps (middle, bottom) removes it, and the remaining denoising steps repair the seams. Native FD 87.2 / 36.9 / 34.1.

before taking the scheduler step (Figure 3 shows the effect). The projection is thus a hard constraint applied to the model’s own belief, not a post-process: the steps that follow see the quantised state and repair the seams it creates, which is why doing this inside the sampler beats quantising the finished image (31.8 vs. 41.6 at 16 px, Section 5.2). It adds no network evaluations; k -means on $R^2 \leq 576$ pixels is negligible next to a UNet pass. We use $f = 0.5$ throughout and report k per resolution; k can also be drawn per sprite from the corpus histogram, which we ablate. The restriction to late steps matters less than the projection does: running it for the whole schedule ($f = 1$) costs two points (33.7 against 31.6), so a reader who wants the simplest possible version can drop f and lose little.

4.3 GUIDANCE RULES WE COMPARE

All guidance rules combine the target-resolution prediction $e_{\text{strong}} = \epsilon_{\theta}(x_t, t, c, b_R)$ with a reference e_{ref} on the same state, $\tilde{e} = e_{\text{ref}} + w(e_{\text{strong}} - e_{\text{ref}})$. CFG (Ho & Salimans, 2022) uses the empty caption. ICG (Sadat et al., 2025b) replaces the caption embedding by i.i.d. Gaussian noise with the embedding’s standard deviation, redrawn at every step, which needs no unconditional training and, as we show, is the rule that survives the change of protocol. A bucketed model also offers a reference with no extra weights—the same network under a *lower* bucket label, optionally through an early EMA snapshot (the *composed* variant)—and that reference can be internalised into the weights with guidance-free training (Chen et al., 2025) so that one forward pass reproduces the two-pass rule. These cross-resolution variants are the ones that win under the biased protocol and fail under the native one; Appendix B gives their equations and their full tables.

5 EXPERIMENTS

5.1 PROTOCOL

For every configuration we draw one sample for each of 3,000 held-out captions, the recaptions of 3,000 held-out sprites, so the prompt distribution matches the data. Our metric is the Fréchet distance of DINOv2-small CLS features (Oquab et al., 2024; Stein et al., 2023) (FD; lower is better), computed at the native resolution R : images are composited on white and nearest-neighbour up-



Figure 4: Eight prompts from the native ground-truth set (sprites drawn at 16 px and excluded from training). Rows: ours with ICG and palette projection, ours with ICG only, ours with CFG 1.5, and the real sprite. Unlike Figure 1, the bottom row here is pixel art rather than a downsampled illustration, which is what the model is being asked to match.

sampled to 224 px. We report it under two reference sets. The *native* reference uses only sprites that are natively at most R px (Section 3); its real-vs-real floor is 4.47 at 16 px. The *biased* reference is the conventional one, a random sample of the corpus, of which 92.9% is downsampled artwork at 16 px; its floor is 3.45. Seed-to-seed standard deviation of our headline configuration is 0.49 over three seeds, and an independent re-training of the same recipe lands within 2.5 FD of the first (20.33/154.30 versus 20.31/151.85 at 20k steps, biased/native), so we do not interpret differences below that scale. We also report CLIP ViT-B/32 image-text similarity (100 cos) and retrieval R@1 of the own caption among 100.

Does the ranking depend on the encoder? A reference set we defined and a feature space we chose could between them manufacture the result, so we recomputed the main comparison with Inception-v3 clean-FID and KID (Parmar et al., 2022) on the same saved samples and the same native reference. The ordering is unchanged — projection 23.79, $k=6$ 23.28, no projection 26.27, CFG 29.37, one global palette 36.05, median filter 66.63, real held-out 10.54 — so the controls still fail and the conclusion is not a property of DINOv2. The separation, however, collapses: what is a $2.8\times$ gap between projecting and not projecting under DINOv2 is $1.10\times$ under Inception, and $k=4$ and $k=6$ swap places. This is what we would expect from an encoder whose input is resized to 299 px: at 16 px the hard edges and four-colour interiors that the projection creates are mostly destroyed by that resampling. We therefore read Inception as a check on the ranking only, and not as independent confirmation of the size of the effect.

Leakage control. Two forms of contamination are possible here and we check both. First, the prompt set must come from the list the training run actually excluded: building it from the evaluation split instead puts training sprites into the prompt set, and the model then reconstructs them almost pixel for pixel. Second, exact-duplicate removal is not enough — sprite sheets contain animation frames, so a held-out sprite can have a near-identical twin in training. Hashing every sprite by an 8×8 luminance-plus-alpha average hash, 1,241 of the 3,000 held-out prompts (41.4%) have such a twin. Rescoring on the 1,759 prompts that do not (Table 1) moves every number by less than 3 FD and changes no ordering, so we report the full set in the main tables and the clean subset here.

A second evaluation set makes the ground truth itself native pixel art: the 210 held-out sprites that are natively ≤ 16 px and were excluded from training (124 after removing near-duplicate twins; Figure 4). Scores there are higher for everyone, including the real sprites, because the prompts describe genuine low-resolution art rather than downsampled illustrations: CFG 1.5 97.7, ICG $w=2$ 76.2, ICG + palette 61.1, real held-out sprites 34.4 (all $n = 124$). The ordering is the same as in Table 3.

Table 1: Removing every prompt whose sprite has a near-duplicate twin in training. Native protocol, 16px.

Configuration	all 3,000 prompts	clean 1,759
v7r, CFG 1.5 (original targets)	167.2	164.2
v8f2, CFG 1.5	123.1	125.0
v8n, CFG 1.5	117.9	120.1
v8n, ICG $w=2$	87.1	87.3
v8n, ICG $w=2$ + palette $k=4$	31.8	29.0

What the reference sprites actually depict. The corpus is cut from sheets on alpha connected components, so besides sprites it contains letters, UI strips, tile fragments and pieces of larger drawings; the colour and contrast filters of the extraction pipeline do not remove them. We labelled every evaluation image with a vision–language model (20 per numbered grid, classes: object, effect, fragment, glyph, junk). Of the 3,000 held-out sprites, 73.2% are standalone objects, 15.7% are fragments, 7.8% are effects, 2.0% are glyphs and 1.3% is junk; in the native reference set the object share is 61.7%. A third protocol therefore keeps only objects on both sides: 825 reference sprites and 2,181 prompts. The ordering is the same as under the other two references (Table 2), which is why we keep the full native protocol for the main tables and report this one as a content control.

Table 2: Object-only protocol at 16px: the reference and the prompts are restricted to sprites a vision–language model labels as standalone objects.

Configuration	FD (object reference) ↓
v7r, CFG 1.5 (original targets)	201.3
real held-out sprites (mixed provenance)	161.3
v8n, CFG 1.5	144.9
v8n, ICG $w=2$	103.0
v8n, ICG $w=2$ + palette $k=4$	37.1

5.2 MAIN RESULTS AT 16 PX

Table 3 separates the three things that changed. Rebuilding the targets moves the model from 167.2 to 117.9 under the native protocol without touching the sampler. Independent-condition guidance then takes it to 87.1, and palette projection to 31.8 ± 0.5 — a larger step than the entire distance between the guidance rules. Projecting inside the sampler is what makes the last step work: quantising the finished samples to the same four colours reaches only 41.6, because nothing repairs the seams that quantisation introduces.

Ablation: data versus sampler. Applying the final sampler to the model trained on the original targets already gives 42.9, so most of the gain is on the sampling side; the rebuilt targets then remove a further 26% ($42.9 \rightarrow 41.6$ with the bucket policy alone, and $\rightarrow 31.8$ once the new sprites are added). The two are complementary: palette projection on top of plain CFG reaches only 58.4, against 31.8 with ICG, so the guidance and the projection do different work.

Is it the palette, or just less information? Two controls replace Π_k at exactly the same steps. The first projects onto *one* four-colour palette shared by every sprite, learned by k -means over the opaque colours of the reference set: the same colour count, without the per-sprite structure. The second applies a 3×3 median filter, which removes the same few-level dither but imposes no palette at all. Neither reproduces the effect (Table 4): both are *worse* than not projecting, so the gain is not “fewer colours” and not “smoother”, but each sprite’s own small palette.

Palette size. Sweeping k at 16px gives 38.1 / 31.8 / 33.7 / 36.9 / 50.6 for $k = 3/4/5/6/8$, and 39.9 when k is drawn per sprite from the corpus histogram. The FD optimum is $k=4$, which is below the median palette of the real data, and it is worth being precise about what that means. Measured on the samples themselves, $k=6$ is the setting that reproduces the medium: six opaque colours and a

Table 3: 16 px, 3,000 held-out captions. *Native*: reference drawn only from sprites natively ≤ 16 px (floor 4.47). *Biased*: the conventional corpus-sample reference (floor 3.45), 92.9% of which is downscaled artwork. The two columns rank the same methods in opposite orders; Section 6 explains why. Palette rows use k -means projection from $t/T \leq 0.5$; “post-hoc” quantises the finished samples instead.

Model (targets)	Sampler	Native FD $\downarrow$	Biased FD
v7r (original targets)	CFG 1.5 (best weight)	167.2	11.1
v8f2 (policy fixed)	CFG 1.5	123.1	14.8
v8n (policy + new data)	CFG 1.0	113.2	19.0
v8n	CFG 1.5	117.9	13.6
v8n	CFG 2	131.1	12.2
v8n	cross-resolution label reference, $w=1.5$	114.9	14.2
v8n	composed reference, $w=1.5$	117.0	12.8
v8n	internalised (1 NFE), $w=1.5$	170.3	10.4
v8n	ICG $w=1.5$ (Sadat et al., 2025b)	87.2	16.2
v8n	ICG $w=2$	87.1	20.7
v8n	ICG $w=1.5$ + post-hoc 4-colour quantisation	41.6	111.0
v8n	ICG $w=1.5$ + palette projection, $k=4$	34.1	111.0
v8n	ICG $w=2$ + palette projection, $k=6$	36.9	71.1
v8n	ICG $w=2$ + palette projection, $k=4$	31.8 ± 0.49	111.2

Table 4: Controls for palette-projected sampling. 16 px, v8n, ICG $w=2$, 3,000 prompts, projection from $t/T \leq 0.5$ in every row.

Late-step treatment of $\hat{x}_0$	Native FD $\downarrow$	Biased FD
none	87.1	20.7
per-sprite k-means, $k=4$ (ours)	31.6	111.2
one corpus-wide 4-colour palette	120.1	233.9
3×3 median filter (de-dither, no palette)	112.2	92.8

flat-neighbour ratio of 0.429 against six and 0.460 for real sprites, where $k=4$ gives four colours and 0.544 — flatter than the real thing. Plain CFG and ICG sit at 52 and 49 colours with a flat ratio of 0.083, and gpt-image-2’s reduced renders at 93 colours and 0.005. The projection therefore moves the samples from far outside the medium to inside it, and the last step from $k=6$ to $k=4$ buys 5 FD by overshooting. We report $k=4$ as the best-FD setting because that is what it is, and $k=6$ as the one a practitioner should use; the disagreement between them is small but it is real, and it is the metric’s, not the method’s.

Is it the network? Every number so far comes from one convolutional network, so the projection could in principle be repairing a failure mode particular to it. We retrained the pilot with the UNet replaced by a 69.7M-parameter transformer (PixArt-style adaptive norm; the resolution bucket enters through the text context, since that design has no class-conditioning path), leaving the corpus, the noise schedule, the sampler and the evaluation untouched. At 20k steps the transformer is the weaker model by a wide margin — 185.6 against 149.1 native FD under plain CFG, a gap of 36.5 — and yet under the reported configuration the two land at 42.4 and 43.1 (two seeds each, spread 0.3 and 0.4), a residual gap of 0.7 with the sign reversed. The projection is not a property of convolutions, and at this scale it matters more than the choice of backbone: it absorbs 98% of the difference between two architectures. We draw no conclusion about which backbone is better — two seeds at 20k steps cannot support that — only that the method transfers, and that what separates the two networks is almost entirely something the decoder can repair.

The cost in prompt alignment. Palette projection trades a little text faithfulness for a large distributional gain: CLIP 100 cos / R@1 are 27.60 / 22.3% for ICG $w=2$ alone, 27.38 / 21.0% at $k=6$ and 27.07 / 17.9% at $k=4$ (real sprites: 19.68 / 1.1%). We therefore report $k=4$ as the best-FD setting and $k=6$ as the balanced one. The cost shrinks as the canvas grows and the quantisation error

with it: at 20 px the projection costs 0.4 points of retrieval (25.3% to 24.9%) and at 24 px 0.9 points (26.2% to 25.3%).

5.3 WHAT THE CORPUS CAN AND CANNOT BUY

The corpus contains material that is not a sprite, material that is badly drawn, and less real pixel art than we would like. We tried to fix all three at training time; none of the three earns a place in the final model. The third attempt is worth reporting at length anyway, because of how it failed.

Content filtering. A vision–language model labels every sprite as object / effect / fragment / glyph / junk (Section 3). Dropping the 8,818 sprites it calls fragment, glyph or junk (training rows 86,290 → 73,845) improves a 20k-step run from 165.6 to 156.7 native FD at equal steps, a 5.4% gain. That is real but small next to the effect of the sampler, and it costs a full retraining, so we report it as an ablation and keep the unfiltered model.

More clean data, and why a short pilot cannot tell you whether it helps. The obvious remedy for a corpus that is only 7.1% natively small is more natively small sprites. We found two sources: the Kenney CC0 packs (7,546 sprites cut from 19 packs, 6,566 usable rows after content filtering, a median palette of three colours — flatter material than anything else we have, but only 41.8% whole objects) and the OpenGameArt CC-BY-3.0 bundle, which our rebuild had never touched (21,722 sprites against which exact pixel hashing finds 13 duplicates, 11,988 usable rows, 82.4% whole objects). Together they take training from 86,290 rows to 118,378.

At 20k steps the addition looks decisive. The Kenney packs alone move the reported configuration from 54.2 to 53.2; adding the CC-BY-3.0 bundle moves it to 42.9, a 19.4% gain and twenty times the seed spread. The natural reading — ours at the time — is that the projection supplies flatness, which the Kenney packs duplicate, while whole objects supply the shape and subject matter it cannot, so data and decoding are complements.

Trained to the same 80k steps as every other model in this paper, the enlarged corpus is *worse*: 39.3 ± 0.3 against 31.8 ± 0.4 over three seeds each, a 24% regression, with plain CFG unchanged (119.7 against 117.9). The ordering at 20k steps is the reverse of the ordering at 80k, and only the second one is the model anybody would ship. We do not offer a mechanism, because the obvious one does not survive measurement: if the new material diluted the 16 px bucket with larger art we would expect its natively-small fraction to be much lower, and it is 24.0% against 28.6% for the material already there — lower, but nowhere near enough to explain a 24% regression.

We report this because it is the second time a 20k-step comparison inverted by 80k in this work; the first was model capacity, where a $2.2\times$ larger network led at 20k (46.1 against 54.2), had lost by 40k (38.3 against 35.8) and was clearly behind at 80k (38.5 against 31.8). Short pilots are how this area is usually made affordable, and in this setting they are safe only for rejecting a change, not for accepting one. Every headline number in this paper is measured at full length for that reason.

One further negative belongs here. Wanting still more material, we processed the one OpenGameArt licence bundle the rebuild had skipped, CC0. It cuts into 54,781 sprites, of which 1,017 are not already in the corpus and 166 are natively 24 px or smaller: it is the bundle the original corpus was built from.

Quality conditioning, and what it does to FD. The same model rates each sprite’s craft from 1 to 5; on the original corpus 25% of the images are rated 1–2 and 8% are rated 5. Prepending the rating to the caption as a tag (“crisp detailed sprite,” / “clean sprite,” / nothing / “rough sprite,”) and asking for the top tag at sampling time is the standard aesthetic-tag recipe and needs no architecture change. It works as intended in the only sense a person can check: the samples are visibly more finished (Figure 5). At full length the numbers repeat the 20k picture — 34.2 untagged and 45.6 with the top tag, against 31.8 for the unconditioned model on the same sampler. Both distributional metrics get worse — $158.4 \rightarrow 208.4$ against the native reference, and $186.5 \rightarrow 227.5$ against a reference restricted to the 719 sprites rated 4–5 — because conditioning on the best 8% of the corpus narrows the output distribution, and FD measures distance to the whole distribution, not to its best part. We keep the main model unconditioned, report the knob for practitioners, and note the disagreement as

Table 5: External comparison, 200 held-out captions, native protocol. Every row is scored after the same k -colour post-process ($k = 4/8/6$ at 16/20/24 px); our row applies it inside the sampler instead. “Real (mixed)” are held-out sprites of the original, mostly downsampled corpus — the targets the biased protocol treats as ground truth.

System	16 px	20 px	24 px
Ours (v8n, ICG + palette projection)	50.8	152.5	192.2
Real sprites (mixed provenance)	77.0	165.5	150.7
gpt-image-2 + downscale	134.4	448.6	478.0
FLUX.2-klein + pixel-art LoRA	150.1	473.9	472.4
SDXL + Pixel Art XL LoRA	244.2	526.3	527.2

a limit of distributional evaluation for this medium: a reviewer who looks at Figure 5 and a reviewer who reads the table will draw opposite conclusions, and both are reading their instrument correctly.



Figure 5: Quality conditioning at 16 px: same model, same prompts and seed, sampled with (top) and without (bottom) the top craft tag. The tagged samples are more finished and more detailed; both FD protocols rate them worse, because the tag narrows the distribution.

5.4 OTHER RESOLUTIONS

The same recipe transfers, and the useful palette size is not simply the data’s median: FD prefers a coarser palette where the canvas is large enough for dither to matter and a finer one at 12 px, where a sprite has only 144 pixels to begin with. Native FD for v8n with ICG $w=2$, three seeds at the best k of each resolution: at 12 px 143.2 ± 0.8 ($k=8$; $k=6$ 143.3, $k=10$ 145.2, $k=5$ 150.7, $k=4$ 156.5, no projection 155.7, CFG 1.5 167.3); at 16 px 31.8 ($k=4$; $k=6$ 36.9, no projection 87.1); at 20 px 96.7 ($k=8$; $k=6$ 99.1, $k=4$ 110.7, no projection 117.0); and at 24 px 124.2 ($k=6$; $k=4$ 128.4, $k=8$ 130.1, no projection 186.8, CFG 1.5 229.2). The gain from projection is largest at 24 px, where the model has the most room to dither, and smallest at 12 px, where it has the least. The same ordering holds one rung above our target range: at 32 px, CFG 1.5 scores 139.2, ICG $w=2$ 97.1 and ICG with an eight-colour projection 74.5 (twelve colours: 78.7), so nothing about the recipe is specific to 12–24 px.

5.5 EXTERNAL SYSTEMS

We compare against three publicly available pipelines on 200 held-out captions: gpt-image-2 and FLUX.2-klein-4B (Black Forest Labs, 2026) with the Limbination pixel-art LoRA (Limbination, 2026) and SDXL with Pixel Art XL (nerijs, 2023), each generating at its native size and then reduced to the sprite canvas by the same cutout and downsampling code we use for real data. To keep the comparison honest we give every system the same palette post-process that our sampler applies internally.

Without the post-process the external systems score 284.4 / 318.9 / 436.4 at 16 px, so the palette step helps them as well; they remain 2.6–4.8 $\times$ further from real pixel art than our model at every resolution. At 16 and 20 px we are also closer to native pixel art than the mixed-provenance sprites

that the conventional protocol calls ground truth, which is a statement about that corpus rather than about our model. At 24 px the real sprites are still ahead of us (150.7 versus 192.2).

The rule as a post-process on other systems. Quantising a finished external sample helps but cannot repair the seams quantisation leaves, the same limitation we measured on our own samples (41.6 post-hoc against 31.8 in-sampler). The projection can nevertheless be applied to another system’s output, by re-noising its sprite to a fraction σ of the schedule and denoising it with our model under the reported configuration. Table 6 reports both halves of what that does: how much closer to real pixel art the result gets, and how much of the original sprite is still there. The second half matters because at $\sigma=1$ the procedure is simply our own model, and a distributional score alone cannot tell repair from replacement. We measure retention in pixel space — alpha-mask IoU and mean RGB distance against the input — because the DINOv2 cosine is saturated at this size: two unrelated sprites for the same caption already score 0.918, against 0.94 for a refinement, so it cannot separate the two cases at all.

At $\sigma=0.6$ the silhouette is about 80% preserved, well above the 0.46–0.53 floor of a freshly generated sprite for the same caption, and native FD improves by 36–41% for all three systems. Sweeping σ on gpt-image-2 gives a monotone trade-off with no knee — 134.4 / 130.2 / 124.2 / 115.9 / 98.7 / 86.5 / 68.0 / 53.6 native FD against retention 1.00 / 1.00 / 0.98 / 0.93 / 0.86 / 0.79 / 0.64 / 0.52 at $\sigma = 0$ to 1 — and the endpoint is a self-check: at $\sigma=1$ the input is discarded entirely and the run reproduces our own samples (53.6 against 50.8, retention 0.518 against 0.513). The modal-colour fraction stays between 0.433 and 0.453 across the whole sweep, against 0.448 for real sprites, so no setting flattens the subject away. The decoding rule is therefore usable on top of a generator we did not train, with σ as an explicit dial between keeping the input and matching the medium. It does not close the gap to generating natively (86.5 against 50.8 for gpt-image-2), which is the honest limit of a post-process: it can fix how a sprite is drawn, not what was drawn (Figure 6). Text alignment makes the same point from the other side: caption retrieval is 22.0% for gpt-image-2’s own sprite, 18.5% after refinement at $\sigma=0.6$ and 15.0% for our model generating the caption itself, with real held-out sprites lowest at 12.5%. Refinement keeps most of a large generator’s semantic advantage while moving the style towards the medium, and at $\sigma=1$ it lands on our own numbers in all three measurements — FD 53.6 against 50.8, retention 0.518 against 0.513, retrieval 16.0% against 15.0% — which is the self-consistency check for the whole procedure. The effect is not particular to 16 px: on gpt-image-2 at 20 and 24 px, $\sigma=0.6$ takes native FD from 448.6 to 265.7 and from 478.0 to 299.0 (41% and 37%), with retention rising to 0.840 and 0.875 as there are more pixels for the same σ to leave alone. The gap to generating natively widens with resolution, though — 265.7 against our 152.5 at 20 px and 299.0 against 192.2 at 24 px, where at 16 px it was 86.5 against 50.8 — which is the same limit seen from further away: a post-process repairs how a sprite is drawn, and the larger the canvas the more of the distance is structure it cannot touch.

Table 6: The decoding rule applied to external sprites, 200 captions, 16 px. σ is the fraction of the schedule re-run. Retention is measured against the system’s own output; the floor column is our model generating the same captions from noise, i.e. what “no retention” looks like.

System	Native FD ↓			Retention at $\sigma=0.6$		Floor
	post-hoc	$\sigma=0.4$	$\sigma=0.6$	IoU	RGB-L1	IoU
gpt-image-2	134.4	115.9	86.5	0.788	45.7	0.513
FLUX.2-klein + LoRA	150.1	116.1	88.2	0.803	44.3	0.530
SDXL + Pixel Art XL	244.2	197.2	155.0	0.813	52.5	0.464
Ours, generated natively		50.8			—	

6 WHY THE TWO PROTOCOLS DISAGREE

Table 3 contains two rankings of the same samples. Under the biased reference the internalised cross-resolution model is best (10.4) and our final configuration is worst (111.2); under the native reference the order is reversed (170.3 versus 31.8). Only one of these references is a description of the medium, but both are the kind of reference set that a paper in this area would plausibly build, so it is worth being precise about what separates them.



Figure 6: Eight captions drawn at random from the 200. Top: gpt-image-2 rendered large and reduced, with the same palette post-process as every other row. Middle: the same sprite re-noised to σ of the schedule and denoised by our model with ICG and the four-colour projection. Bottom two: our model generating the caption from noise, and the held-out sprite. The refined sprites keep the subject — the aggregate agrees, with a modal-colour fraction of 0.453 against 0.448 for real sprites — while the colours and edges move towards the medium. They also look better than our own samples for the same captions, which score 86.5 and 50.8 native FD respectively: what our model still lacks is not the decoding rule but the ability to draw the subject.

The two references are different distributions. At 16 px the biased reference is 92.9% artwork drawn at 25–64 px and box-downsampled. Downsampling averages neighbouring pixels, so those images have soft transitions and large palettes: their median opaque-colour count is 32 against 6 for sprites drawn at 16 px, and their fraction of exactly-equal adjacent pixel pairs is 0.30 against 0.46. Scoring against them rewards a generator that produces the same softness. This is not a subtle effect: real held-out sprites of mixed provenance score 136.1 against the native reference (77.0 after palette quantisation), i.e. the “ground truth” of the conventional protocol is itself far from native pixel art.

What guidance does to those statistics. Per-image statistics on 600 samples explain the reversal. Classifier-free guidance sharpens by increasing contrast, not by reducing the palette: at $w = 1.5$ it emits 56.2 colours with contrast 59.4 (real: 11.0 colours, contrast 62.3), and raising w further increases both contrast and colour count, which the biased reference rewards (12.2 at $w=2$) and the native one punishes (131.1). ICG moves all three statistics towards the data at once (50.9 colours, flat-pair fraction 0.381, contrast 63.9), which is why it is the one external rule that improves both protocols’ agreement and composes with the projection. The cross-resolution reference, by construction, is a *lower-resolution* belief: it is smoother than the target-resolution prediction, so extrapolating away from it adds contrast in the same way CFG does while leaving the palette untouched, and internalising it bakes a smoothed reference into the weights — which is why the internalised model is worse than its own base model under the native reference even at $w = 1$ (178.8 versus 113.2), before any extrapolation is applied.

What a sprite generator is asked to produce. It is tempting to read the external systems’ more attractive samples as better generation, but the target here is not an attractive picture. A sprite sheet is a production asset: our content labels put 73.2% of the held-out sprites in the object class, 7.8% in effects (fire, smoke, sparkles, projectiles) and 15.7% in fragments — tiles, strips, borders and parts of larger drawings that a game assembles at run time. At 16×16 an explosion particle or a wall tile is a small cluster of flat colours; drawing it as a polished illustration with shading and highlights is

the wrong answer, however pleasant it looks in isolation. This is the sense in which the medium, not the eye, sets the objective, and it is what the native protocol measures: a system that renders every caption as a miniature painting scores 134–527 against real low-resolution art, while our samples, which are often plain, score 50.8–192.2.

It also explains the quality-conditioning result of Section 5.3 at the level of mechanism. The craft-5 sprites in the corpus are predominantly finished items and characters; conditioning on that tag does not only remove badly drawn sprites, it removes whole legitimate categories — effects and tiles — from the output distribution. The samples look better one at a time and the distributional score gets worse, because the model is now answering a narrower question than the one the corpus asks.

Consequence for practice. A benchmark built from “sprites” that are downsampled illustrations will keep selecting methods that produce downsampled illustrations. We do not claim the native protocol is the only defensible choice — a corpus of 25–64 px artwork is a legitimate target if that is what one wants to generate — but it should be a deliberate choice, and papers should say which one they used. We report both throughout, and we release the split so that either can be reproduced.

7 LIMITATIONS AND CONCLUSION

Where the remaining error lives. Our samples are recognisable but crude, and it is worth saying which knobs we ruled out. Swapping the frozen CLIP ViT-B/32 text encoder for ViT-L/14 (512 to 768 dimensions) changes nothing at equal steps: 164.0 against 165.6 native FD, with retrieval R@1 slightly *lower* (19.6% against 20.6%). Training longer is exhausted: continuing the main model past 80k gives 31.8, 30.8 and 32.3 at 80k, 100k and 120k steps on the final recipe, a flat line inside seed noise. Capacity does not help either at this data scale — a $2.2\times$ model (72.5M to 161.2M parameters, same data and schedule) leads early and loses late: 46.1 against 54.2 at 20k, 38.3 against 35.8 at 40k, and 38.5 against **31.8** at 80k. The one intervention that visibly improves the samples is quality conditioning, which narrows the distribution and therefore scores worse.

What is left is the corpus. It holds roughly 50k sprites, a vision–language model rates 25% of them 1–2 out of 5 for craft, and 19% of the held-out set is not a standalone object at all. A model fitted to that mixture reproduces the mixture, which is the correct behaviour under our own protocol and also the reason we do not present this as a system for producing attractive sprites. The path we would take next is more and better data — curated sprite collections, or a much larger corpus with the craft rating used for filtering rather than conditioning — not a larger network.

Limitations. (i) *Scope.* Everything is measured on models we trained, on one family of game-sprite corpora; the transformer run rules out convolutions as the explanation but not a shared inductive bias of pixel-space diffusion, and we have not tested a second domain. (ii) *Metric.* FD with DINOv2 on 12–24 px images upsampled to 224 px assigns one sprite pixel to one or a few patches, and is sensitive to exactly the colour and contrast statistics that our projection changes; we report CLIP alignment, both references and a second feature space throughout, but the second feature space is much less sensitive at this size and we have no human study, which remains the obvious next check for a claim about a medium people look at. (iii) *Palette tuning.* k and the projection start f were selected on the same held-out FD we report. Splitting the 3,000 captions into halves by index and choosing on one of them selects the same $k=4$, which on the disjoint half scores 33.36 against 38.63 / 35.62 / 38.27 / 43.76 for $k = 3/5/6/8$ (values at $n=1500$ are not comparable to the $n=3000$ numbers elsewhere, since FD is biased by sample size), so that choice transfers. For f the honest statement is that the optimum is flat rather than validated: $f = 0.4$ and 0.5 give 31.4 and 31.6, within the 0.49 seed spread, and each wins on one half, while every value outside that interval is worse (32.9 at 0.3, 32.3 / 32.8 / 32.9 at 0.6/0.7/0.8, 33.7 when the projection runs for the whole schedule). Anything in $[0.4, 0.5]$ will do; the late-step restriction is worth about two points, so most of the effect is the projection itself rather than its timing. The per-sprite variant drawn from the corpus histogram (39.9) is still worse than the tuned constant, so some of the gap is fitted to the metric. (iv) *Where we still lose.* At 24px real sprites remain closer to native pixel art than our samples (150.7 versus 192.2), and on individual prompts a large generator plus downscaling still produces a more attractive picture than ours even though it is further from the medium — our samples inherit the crudeness of the data as well as its grammar. (v) *Alignment.* The strongest setting costs 4.4

points of CLIP retrieval; at $k=6$ the cost is 1.3 points, and prompt following at these sizes is weak for every model we tested.

Conclusion. For 12–24px sprites, what the model is asked to imitate matters more than which guidance rule it uses. The corpus that this literature trains on is 93% downsampled artwork at 16px, it leaks 15.5% of its own held-out set through exact duplicates, and the evaluation reference built from it rewards the softness it introduced. Rebuilding the targets and scoring against real low-resolution pixel art moves FD-DINOv2 from 167.2 to 117.9; projecting the predicted x_0 onto a small per-sprite palette during the late denoising steps — the property of the medium that continuous diffusion cannot express — takes it to 31.8 ± 0.5 , against 134.4 for the strongest external pipeline under the same palette post-process. The same change of reference set reverses the ranking of our earlier cross-resolution guidance results, which is the part of this paper we would most like others to check.

REPRODUCIBILITY STATEMENT

Section 4.1 and Appendix C specify the architecture, data processing, training recipe and sampler. Equation 3 and Appendix B give the full internalization objective with all hyperparameters. Section 5.1 and Appendix C specify the evaluation protocol: prompt set, chunked seeding, reference set, preprocessing and floors. Appendix E gives the exact settings of every guidance baseline, and Section 5.5 the conversion pipelines for the external systems. We will release code for training, sampling, palette projection and evaluation, the captions, the per-sprite licence manifest of the new material, and the held-out split **[TODO: anonymous code link in the supplementary material]**. The raw sprite corpus carries per-asset licences and will be released as a list of sources after a licence audit (see the ethics statement).

ETHICS STATEMENT

The model generates small game sprites. It adds no capability beyond existing text-to-image systems, and its outputs (12–24px) have little potential for targeted misuse. Its intended application raises the usual concerns of generative art tools, namely the displacement of pixel artists and the reproduction of styles present in the training corpus. The training data are community sprites from OpenGameArt and the Kenney and Liberated Pixel Cup packs. The 14,999 sprites added in this work carry a recorded source zip, member path and licence (CC-BY-4.0, CC-BY-SA-3.0, OGA-BY-3.0); the older material predates that bookkeeping, so a per-asset licence audit precedes any release of the full corpus. Captions were produced by a commercial vision-language model; external baselines were generated through commercial APIs under their terms of use. No human subjects were involved.

AI USE STATEMENT

Generative AI tools were used in this work in the following ways. (a) Inside the method and its data pipeline: vision–language models produced the training and evaluation captions (gemini-3-flash for the original corpus, gemini-3.8-flash for the material added here), labelled every evaluation sprite by content type and rated sprite craft for the ablations of Section 5.3; two commercial image generators served as external baselines (Section 5.5). Each of these uses is an experimental component, is described where it is used, and its outputs were spot-checked against manual inspection — the caption classifier was validated against image-level labels on 2,980 held-out sprites (82.5% agreement), and the content and craft labels were checked on random grids by the authors. (b) In the research process: an LLM-based coding assistant implemented and ran experiments, searched and verified related work, and drafted and edited parts of this manuscript. Every number reported here comes from a script in the released code and was regenerated from the saved samples; the authors reviewed all AI-assisted work, including the negative results, and take responsibility for the final content, its claims and its errors.

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A THE REBUILT CORPUS, STEP BY STEP

Counts. The original corpus holds 37,489 unique sprites (after cutting sheets into connected components at native scale and dropping glyphs, UI and near-monochrome crops). Native sizes: 710 at most 12 px, 2,675 at most 16 px, 9,687 at most 20 px, 12,516 at most 24 px, 19,611 at most 32 px. Exact-pixel hashing finds 4,888 duplicate copies; 15.5% of the 3,000 held-out evaluation sprites and 17.0% of the 200 external-comparison sprites have a pixel-identical twin that the path-based hold-out leaves in training. Integer-upscaled pixel art, which could be restored by dividing by its block size, accounts for only 4.6% of the corpus.

New material. The CC-BY-4.0, CC-BY-SA-3.0 and OGA-BY-3.0 OpenGameArt collections (8.8 GB of 2D art) yield 37,226 cut sprites. Filtering by the same rules removes 8,184; cross-corpus pixel hashing removes a further 6,062 that duplicate the existing corpus; a filename filter removes 367 font, UI and text sheets; average-hash collapse of animation frames removes 3,456; a cap of 60 sprites per source sheet removes the rest, leaving 14,999 sprites from 1,207 sheets. Verification of this set: median palette 6 colours, 98.2% of visible pixels fully opaque, mean flat-neighbour fraction 0.434, native sizes 234 / 1,726 / 5,374 / 6,028 at most 12 / 16 / 20 / 24 px. Every sprite keeps its source zip, member path and licence in a manifest.

Captions. We caption the new sprites with gemini-3.8-flash on numbered 4×5 grids of nearest-upscaled sprites on a checkerboard, one caption per cell, 6–15 words, with an explicit “unclear abstract shape” escape. Of 14,999 sprites, 14,845 receive a caption (99.0%); dropping the escapes, degenerate strings and 11 rows that echoed the instruction leaves 13,987 usable captions, 99.6% distinct, median 13 words. We selected the captioner by running the same grid through ten vision-language models: gemini-3.8-flash parsed all 20 cells in both trials at 22 s per call and was the most accurate, while a larger sibling truncated at cell 11 in both trials and several models mislabelled weapons and buildings.

Training-set composition. With duplicates removed and downscaling limited to $1.5\times$, the rebuilt corpus yields 86,290 training rows (12/16/20/24 px buckets: 7,357 / 11,733 / 15,087 / 17,199), against 64,990 without the new sprites and 180,533 under the original policy, which fed 48 px artwork to the 16 px bucket.

Near-duplicate collapse is harmful. Collapsing average-hash near-duplicates in the *old* corpus as well (37,489 $\rightarrow$ 26,504 groups) costs more than it gains: at 20k steps it scores 21.07 biased / 162.23 native against 20.31 / 151.85 for the same recipe without the collapse. We therefore drop only exact duplicates in the main corpus and apply the collapse only to the new material, where single sheets contributed up to 563 frames of one character.

B CROSS-RESOLUTION GUIDANCE: EQUATIONS AND FULL TABLES

The cross-resolution rules referenced in Section 4.3 use the guided prediction $\tilde{e} = e_{\text{ref}} + w(e_{\text{strong}} - e_{\text{ref}})$ with

$$e_{\text{ref}}^{\text{label}} = \epsilon_{\theta}(x_t, t, c, b_{\text{low}}), \quad e_{\text{ref}}^{\text{comp}} = \epsilon_{\theta_{\text{snap}}}(x_t, t, c, b_{\text{low}}), \quad b_{\text{low}} < b_R, \quad (2)$$

i.e. the same network (or an early EMA snapshot of it) queried under a lower resolution bucket, with $b_{\text{low}} = 12$ at 16 px and $b_{\text{low}} = 16$ at 20 and 24 px. The internalised variant follows guidance-free training (Chen et al., 2025): the network gains a scalar input $\beta \in (0, 1]$ and is trained with

$$\hat{e}_{\beta}(x_t, t, c, b) = \beta s_{\theta}(x_t, t, c, b, \beta) + (1 - \beta) \text{sg}[\epsilon_{\theta_{\text{snap}}}(x_t, t, c, b_{\text{low}}(b))], \quad (3)$$

so that setting $\beta = 1/w$ at inference reproduces Eq. equation 2’s rule in a single forward pass.

On the original targets this was our strongest result: at 16 px the internalised model reached 7.23 against 11.28 for the best-tuned CFG under the biased protocol, 6.25 when combined with ICG, with controls (empty-caption reference 10.05, independent-condition reference 7.96, online reference 9.13) isolating the lower-resolution reference as the source of the gain. On the rebuilt model, and scored against native pixel art, the same rules give 114.9 (label), 117.0 (composed), 178.8 ($\beta = 1$, no extrapolation) and 170.3 ($w = 1.5$), against 113.2 for unguided sampling and 87.1 for ICG. The mechanism is in Section 6: the reference is a lower-resolution belief, so extrapolating away from it adds contrast without reducing the palette, and internalising it bakes a smoothed reference into the weights.

C IMPLEMENTATION AND PROTOCOL DETAILS

Table 7: Training and sampling configuration. v7h and v7r differ only in the caption files.

Denoiser	diffusers UNet2DConditionModel, 4 input/output channels (RGBA), widths 128/256/512, 2 layers per block; down: CrossAttn, CrossAttn, plain; up: plain, CrossAttn, CrossAttn; 72.5M parameters
Text	frozen CLIP ViT-B/32 text encoder, 77 tokens, last hidden states (512-d) as cross-attention context
Resolution	class embedding over $B = \{12, 16, 20, 24, 32, 48, 64\}$ added to the timestep embedding; never dropped
Diffusion	ϵ -prediction, $T = 1000$, cosine (squaredcos_cap_v2) schedule
Base training	from scratch, 80k steps, AdamW, constant lr $5 \cdot 10^{-5}$, EMA 0.999, caption dropout 10%
Batches	one bucket per batch, bucket drawn proportionally to its row count; batch 320 / 256 / 224 / 192 / 128 / 64 / 40 for 12 / 16 / 20 / 24 / 32 / 48 / 64 px
Snapshot	10k-step EMA of an independent 10k-step run with the same recipe (v7r); step-10k EMA of the run itself (v7h)
Internalization	init v7r, 10k steps, same optimizer/lr/EMA/batches; $\beta \sim \mathcal{U}(0.4, 1)$, $\beta=1$ w.p. 0.25 and for 12 px rows; reference rung 12 (for 16 px rows) and 16 (for 20 and 24 px rows); β via 256-d sinusoidal embedding of 1000β and a zero-initialized linear map into the timestep features (32,768 parameters)
Sampling	100-step ancestral DDPM, EMA weights, one sample per prompt

Canvas preprocessing. A sprite larger than the target side R is reduced with a premultiplied-alpha box filter, preserving aspect ratio; a sprite at most $R/2$ is enlarged by an integer nearest-neighbour factor. It is then centred on a transparent $R \times R$ canvas, and alpha is thresholded at 128 (pixels below become fully transparent, with RGB cleared). Generated samples are stored with the same hard alpha. The same function produces the FD reference set, the held-out floor set and the “downscale” external baselines.

FD. Real sprites are split once with a fixed shuffle into a 3,000-sprite reference set and a disjoint held-out pool. The first 3,000 held-out sprites supply the floor set and, through their recaptions, the 3,000 evaluation prompts. Each image is composited on white and resized to $R \times R$ (box filter; generated images are already $R \times R$). It is then nearest-neighbour upsampled to 224×224 , normalized with ImageNet statistics, and embedded by the CLS token of `facebook/dinov2-small`. FD is the Fréchet distance between Gaussian fits of the two feature sets (3,000 vs. 3,000; 200 vs. 3,000 for the external protocol). Samples are generated in chunks of 1,000 prompts, and chunk i uses seed $s + i$.

CLIP. The same saved samples are composited on mid-grey, nearest-neighbour upsampled to 224px and scored with CLIP ViT-B/32 against the caption they were sampled from. We report $100 \cos$ and $R@1$, the fraction of samples whose own caption ranks first among itself and 99 random held-out captions (chance 1%).

D RECAPTIONING

Table 8: Caption statistics before and after recaptioning (43,342 unique training and evaluation images).

	BLIP (v7h)	gemini-3-flash (v7r)
Distinct captions	23.5%	98.6%
Words per caption (median / mean)	8 / 7.6	12 / 12.1
Occurrences of the most frequent caption	7,669 (“a pixel art sprite”)	6 (“human”)
Share of data covered by the ten most frequent captions	$\approx 25\%$	$< 0.1\%$

Images were sent as numbered grids on a checkerboard (20 per request, 12 for retries). For each cell, the instruction asked for a caption of 6–15 words that names the object first and then its distinguishing colours, materials and parts, describes only what is visible, and avoids uninformative words (“pixel art”, “sprite”, “8-bit”, “icon”). Of the 43,342 unique images, 41,560 received a caption and 41,103 (94.8%) were usable. The remainder were declared an “unclear abstract shape” by the captioner and keep their BLIP caption. Across the three caption files (49,652 rows), 95% of rows were replaced. The total cost was about 18.7M tokens. The held-out evaluation prompts are the recaptions of the held-out sprites, so the evaluation prompts follow the training caption distribution.

E GUIDANCE BASELINES: EXACT SETTINGS

All baselines replace CFG in Eq. equation 2, on the same model, sampler, prompts and initial noise. Random draws for ICG and TSG come from a separate generator, so that switching methods does not change the sampler noise.

- **CDG** (Han et al., 2026), $R_{\text{deg}} = 1$: every position covered by the tokenizer’s attention mask (BOS, words, first EOS) takes the embedding of the empty caption at that position, while the remaining padding positions keep the caption’s states; the bucket is unchanged. On our recaptions, on average 16.3 of 77 positions are content, and no prompt fills all 77 (for such a prompt CDG would equal CFG).
- **ICG** (Sadat et al., 2025b): $\hat{c} = \text{std}(c) \cdot n$ with $n \sim \mathcal{N}(0, I)$ of the caption embedding’s shape, redrawn at every step; the target bucket is kept. **ICG-label**: the caption is kept and the bucket is drawn uniformly from $\mathcal{B}$ at every step.
- **TSG** (Sadat et al., 2025b): the time embedding (before the bucket embedding is added) is perturbed by $s(t/1000)^\alpha \text{std}(\cdot) n$ in all layers. “SD” setting: $s = 3$, $\alpha = 0.25$, applied for $t \geq 400$; “DiT” setting: $s = 2$, $\alpha = 1$, all t , $w = 2.5$.
- **SEG** (Hong, 2024): self-attention queries are Gaussian-blurred over their square layout (kernel size $\lceil 6\sigma \rceil + 1$, truncated to the map, reflect padding; $\sigma = \infty$ replaces the queries by their spatial mean), in the mid block (“mid”), in the down/up blocks at $R/2$ (“d1+u1”) or everywhere (“all”); $w = 3$. The “all, $\sigma = 1$ ” setting gives 122.52.
- **PAG** (Ahn et al., 2024): the self-attention map of the mid block is replaced by the identity.

- **Autoguidance** (Karras et al., 2024): the 10k snapshot under the target bucket and the same caption.

F EXTRAPOLATION-WEIGHT MEASUREMENTS

Table 9: MSE-optimal extrapolation weight w^* on forward-noised data (v7h; 10 timesteps from $t = 10$ to 990; 3,000 sprites; ranges over timesteps) and the relative denoising loss of the guided prediction at $w = 1.5$.

Resolution / rung	Data	label w^*	snapshot w^*	composed w^*	$L(1.5)/L(1)$
16 / 12	held-out	0.88–1.04	0.83–1.07	0.85–1.07	1.02–1.88
16 / 12	training	0.94–1.00	1.00–1.07	0.99–1.07	1.02–1.90
20 / 16	held-out	0.92–1.02	0.88–1.05	0.89–1.06	1.01–1.81
24 / 16	held-out	0.91–1.02	0.85–1.05	0.88–1.04	1.03–1.95

The “label” reference is the final model under the lower bucket, “snapshot” is the 10k snapshot under the target bucket (autoguidance), and “composed” is the snapshot under the lower bucket. All three are evaluated on the same x_t and caption. On self-rolled states (2,000 held-out sprites), a sprite is noised to k steps before a target position in the 100-step schedule (positions 25, 45, 65, 85, 95), and the deployed sampler (bare, or label-guided with $w = 2$) runs k steps. w^* is then measured in x_0 -space against the known sprite. Over all $k \geq 5$ points, the maximum w^* is 0.895 / 1.069 / 0.989 (label / snapshot / composed) at 16 px, 0.825 / 0.997 / 0.978 at 20 px and 0.819 / 0.888 / 0.876 at 24 px. No point reaches 1.3.

G RESULTS ON THE BLIP-CAPTION MODEL v7H

Table 10: 16px results on v7h (BLIP captions, BLIP evaluation prompts) next to v7r. Three seeds where $\pm$ is given, otherwise seed 0.

Method	NFE	v7h (BLIP)	v7r (recaptioned)
CFG $w = 4$ (default)	2	21.52 $\pm$ 0.47	20.18
Best CFG ($w = 1.5$)	2	12.49 $\pm$ 0.72	11.28 $\pm$ 0.54
Autoguidance, 10k snapshot	2	8.73 $\pm$ 0.26	9.78 $\pm$ 0.50
Lower-bucket label reference	2	8.52 $\pm$ 0.29	8.86 $\pm$ 0.06
Composed reference	2	7.53 $\pm$ 0.19	8.50 $\pm$ 0.25
ICG, $w = 1.25 / 1.5$	2	8.88 / 11.29	7.68 / 7.46 $\pm$ 0.37
Internalized, $w = 1.5$	1	7.23 $\pm$ 0.34 (7.63 / 7.02 / 7.05)	7.23 $\pm$ 0.45
Internalized, $w = 1.25 / 2$	1	8.77 / 9.56	–
Internalized ($w = 1.5$) + CFG 1.25 / 1.5; ($w = 1.25$) + CFG 1.5	2	8.07 / 8.16 / 8.34	–
Higher-bucket reference, buckets 20 / 24 / 64, $w = 2$	2	17.14 / 17.27 / 19.34	–

On v7h the internalized model beats the composed rule at every seed (7.63 / 7.02 / 7.05 vs. 7.67 / 7.32 / 7.61). Higher buckets are harmful at every resolution: at 12 px, bucket 16 gives 14.45 vs. best CFG 9.71; at 20 px, buckets 24 / 32 give 46.62 / 59.13 vs. 39.53; at 24 px, bucket 32 gives 101.67 vs. 67.34 (seed 0).

What the lower-resolution belief looks like (v7h, 16 px). Sampling each reference on its own (median unique colours over opaque pixels, fraction of exactly equal adjacent pixel pairs, mean adjacent $|\Delta\text{RGB}|$) gives 78 / .015 / 21.6 for the 12 px belief, against 73 / .030 / 32.0 for the strong model (CFG $w = 4$) and 34 / .202 / 30.2 for real sprites. The 24 px and 64 px beliefs have contrast 31.9 and 40.9. The 12 px belief is thus not a simpler image with fewer colours; it is a lower-contrast rendering of the same layout. A contrast reduction that is *not* structured like the belief does not work: shrinking the strong prediction’s own $\hat{x}_0$ contrast towards its mean and using that as the reference gives 74.8–444.4.

Table 11: Wall-clock time for 1,000 samples at 16 px (A100-80GB, exclusive; 100 DDPM steps; batch 500; including model loading), measured on v7h. The internalized model has the same architecture plus 32,768 parameters and runs at 1 NFE; it was not timed separately.

Configuration	NFE	s / 1,000 samples	peak memory (MiB)
One pass per step (no guidance; internalized model)	1	42.5	4951
CFG	2	73.8	4951
Lower-bucket label reference	2	72.7	4951
Autoguidance / composed reference (snapshot in memory)	2	73.3 / 74.0	5507
SD- π XL (Binninger & Sorkine-Hornung, 2024), one 16 px sprite (shared node)	–	29,500 per sprite	–

H COST

I SDXL SIZE-CONDITIONING PILOT

Table 12: SDXL base 1.0 on the first 1,000 captions of COCO-30k shard 0 (1024 px, Euler 30 steps, identical initial noise). FD with DINOv2 ViT-L/14 CLS (224 px, bicubic) against real images 1,000–2,999; floor (real images 0–999) 507.05; CLIP ViT-L/14 100 cos (real 25.91). “orig” denotes the `original_size` condition of the weak branch.

Setting	FD	CLIP
CFG $w = 3 / 5 / 7$	662.22 / 644.18 / 648.44	26.15 / 26.59 / 26.82
CFG $5 + 1 \times (e_c - e_{c, \text{orig}=512})$	644.13	26.62
CFG $5 + 1 \times (e_c - e_{c, \text{orig}=256})$	658.98	26.46
CFG with <code>negative_original_size 512</code> (common practice)	650.98	26.46
Size reference only, orig 512, $w = 3$ (no CFG)	969.68	21.26

J EXTERNAL SYSTEMS: PIPELINE AND QUALITATIVE COMPARISON

The gpt-image-2 and nano-banana-2 renders were requested at 1024 px through the vendors’ image-generation APIs with the prompt “<recaption>, pixel art sprite of a single object, plain flat white background, centred, no text”. The sprite is cut out by flood-filling, from the image border, all pixels within 18 (per channel, 8-bit) of the median border colour, and then cropped to its content. *Downscale* applies our canvas preprocessing (Appendix C). *PixelOE* pads the cut-out to a square on white, pixelizes the RGB in contrast mode with colour matching directly to $R \times R$, and takes alpha by majority vote of the cut-out mask in each cell. *Make Your Own Sprites* resizes the same square to $4R \times 4R$ (bicubic), applies the official network with 4 px cells, and reduces the output by 4 with nearest-neighbour sampling, with alpha as for PixelOE. We bypass the official script’s minimum input size of 128 px, because it would change the cell count for 12–24 px targets. The SDXL + Pixel Art XL LoRA row uses SDXL base 1.0 with the fused LoRA (scale 1), Euler with 30 steps, CFG 7 and the same prompt and conversions.

K CAPTIONS OF THE QUALITATIVE FIGURES

Held-out prompt indices 3, 7, 12, 20, 33, 41, 55, 67, left to right in Figure 1:

1. Crou (*a caption the vision model could not resolve; kept to show the untouched prompt distribution*)
2. Alien soldier, blue with green armor, lunging forward while gripping a black rifle
3. Top-down character head with brown hair, black goggles, white antenna, and pale skin
4. Platypus like creature with brown body, yellow beak, and large eyes in swimming pose
5. Man: blocky person with short brown hair, red shirt, and dark blue pants
6. Stone brick wall with yellow rectangular bricks and a central red gem
7. Large fire, white-yellow core, intense orange glow, dark red jagged outlines
8. Glowing sun, bright yellow center with an orange border and a white cross



Figure 7: “Generate large, then shrink” vs. ours at 16 px (same captions as Figure 1; $n = 200$ FD in parentheses). The 1024 px renders (shown as downsampled thumbnails) often follow the caption closely. After conversion they keep soft shading, anti-aliased edges and many colours, which are absent from the data. The “downscale” sprites shown here are re-derived from the saved renders with the same deterministic pipeline that produced the scored ones.

Figure 3 uses indices 11, 20, 33, 50, 67 of the same file.

L WHAT IS NOT IN THIS DRAFT

For transparency about scope: SD- π XL (Binniger & Sorkine-Hornung, 2024) is included only qualitatively, because its score-distillation optimisation costs about 8 GPU-hours per sprite and we completed 6 of 30 sprites; the internalised cross-resolution model is reported at 16, 20 and 24 px on the rebuilt corpus but its 12 px variant (which needs an extra 8 px rung) was measured only on the earlier corpus; CLIP alignment is reported at 16 px only; and no human preference study was run, which the limitations discuss.